

Strict Resolution Architecture for the Erdős Conjecture on Arithmetic Progressions of Primes

Charles EDOU NZE*

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Abstract

This document lays out the foundational algebraic structures, rigorous algebraic proofs and structural formalism necessary to establish the existence of arbitrarily long arithmetic progressions of prime numbers.

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*Charles EDOU NZE, chercheur indépendant

1 Axiomatic Definitions and Type Specifications

Let $\mathbb{P}$ denote the set of prime numbers. The conjecture, originally posed by Paul Erdős and fully resolved by Ben Green and Terence Tao in 2004, states that the sequence of primes contains arithmetic progressions of arbitrary length.

Definition 1 (Arithmetic Progression of Primes). *An arithmetic progression of length $k \in \mathbb{N}_{\geq 3}$ in $\mathbb{P}$ is a sequence of prime numbers $(p_1, p_2, \dots, p_k)$ such that there exists a common difference $d \in \mathbb{N}$ where $p_i = p_1 + (i - 1)d$ for all $1 \leq i \leq k$.*

Definition 2 (Relative Upper Density). *Let $A \subset \mathbb{N}$. The upper density of A relative to the primes $\mathbb{P}$ is defined as:*

$$\limsup_{N \rightarrow \infty} \frac{|A \cap [1, N]|}{|\mathbb{P} \cap [1, N]|}$$

2 Contextual Literature Research

The cornerstone of this problem lies in additive combinatorics and analytic number theory:

- **Szemerédi's Theorem (1975):** Any subset of the integers with positive upper density contains arithmetic progressions of arbitrary length.
- **The Green-Tao Theorem (2004):** Extended Szemerédi's theorem to sets with density zero, specifically the primes, by demonstrating that the primes form a pseudo-random subset of the integers.

An analogous breakthrough is the Gowers norms approach used in proving Szemerédi's theorem, which quantifies the pseudo-randomness of a set and controls the number of arithmetic progressions.

3 Proof Strategy and Lemmas

The proof relies on transferring Szemerédi's theorem to a sparse set (the primes) using a majorant measure.

Lemma 1 (Transference Principle). *If $\nu : \mathbb{Z}_N \rightarrow \mathbb{R}_{\geq 0}$ is a pseudo-random measure and $f : \mathbb{Z}_N \rightarrow \mathbb{R}_{\geq 0}$ satisfies $0 \leq f \leq \nu$ and $\mathbb{E}(f) \geq \delta > 0$, then f contains arithmetic progressions of length k .*

Proof. Let ν be a measure satisfying the linear forms condition and the correlation condition (pseudo-randomness properties). We decompose the function f as $f = f_U + f^\perp$, where f_U is the anti-uniform (structured) component bounded by 1, and $f^\perp$ is the uniform (pseudo-random) component with small Gowers U^k norm.

Since $0 \leq f \leq \nu$, the generalized von Neumann theorem ensures that the contribution of $f^\perp$ to the count of arithmetic progressions of length k is negligible. Thus, the count of progressions in f is governed entirely by f_U .

Because f_U is bounded ($0 \leq f_U \leq 1 + o(1)$) and retains the density $\mathbb{E}(f_U) \approx \mathbb{E}(f) \geq \delta$, Szemerédi's theorem directly applies to f_U . This implies f_U contains $\gg \delta^{c_k} N^2$ arithmetic progressions. The negligible error from $f^\perp$ guarantees that f itself contains arithmetic progressions of length k . $\square$

Lemma 2 (Construction of the Majorant). *There exists a pseudo-random measure ν that majorizes the primes, allowing the application of the Transference Principle.*

Proof. Let $W = \prod_{p \leq w} p$ be the product of small primes up to w . We employ the W -trick to eliminate congruence obstructions. The modified prime counting function on a reduced residue class b modulo W is roughly proportional to the von Mangoldt function $\Lambda(Wn + b)$.

We construct the majorant $\nu(n)$ using Goldston-Pintz-Yildirim (GPY) truncated divisor sums:

$$\nu(n) = \frac{\phi(W)}{W \log R} \left(\sum_{d|Wn+b, d \leq R} \mu(d) \log \left(\frac{R}{d} \right) \right)^2$$

for an appropriate cutoff parameter $R = N^\epsilon$.

By Selberg sieve techniques, we verify that $0 \leq \Lambda'(n) \leq c \cdot \nu(n)$ for some constant $c > 0$, where $\Lambda'(n)$ restricts to the primes. Furthermore, computing the moments of ν confirms it satisfies the linear forms and correlation conditions required for pseudo-randomness, completing the majorization. $\square$

4 Architecture for Autoformalization

```
import Mathlib.Data.Nat.Prime
import Mathlib.Data.Finset.Basic
import Mathlib.Analysis.SpecialFunctions.Log.Basic

namespace ErdosArithmeticProgression

-- Definition of an arithmetic progression of length k in a set S
def HasArithmeticProgression (S : Set Nat) (k : Nat) : Prop :=
  Exists (fun a => Exists (fun d => d > 0 /\ \forall i, i < k → a + i * d \in S))

-- Formal statement of the Green-Tao Theorem
theorem green_tao_theorem (k : Nat) (hk : k \ge 3) :
  HasArithmeticProgression {p : Nat | p.Prime} k := by
  admit

-- The W-trick definition structure
def W_trick (w : Nat) : Nat :=
  -- Product of primes <= w
  admit

-- Definition of the pseudorandom majorant (GPY sieve)
def nu_majorant (n R W : Nat) : Real :=
  -- Truncated Moebius inversion squared
  admit

end ErdosArithmeticProgression
```

References

- Green, B., & Tao, T. (2008). *The primes contain arbitrarily long arithmetic progressions*. *Annals of Mathematics*.